

* Introduction to Probabilities -

Probability is defined as the chance of happening or occurrences of an event. Generally, the possibility of analyzing the occurrence of any event with respect to previous data is called probability.

For example, if a fair coin is tossed, what is the chance it lands on the head? These types of questions are answered under probability.

- What is Probability Theory -

It uses the concept of random variables and probability distribution to find the outcome of any situation. Probability theory is an advanced branch of mathematics that deals with the odds and statistics of happening an event.

- Theoretical Probability -

It deals with assumptions in order to avoid unfeasible or expensive repetition of experiments. The theoretical probability for an event A can be calculated as -

$$P(A) = \frac{\text{Number of outcomes favourable to Event A}}{\text{Number of all possible outcomes}}$$

(Number of all possible outcomes)

- Experimental Probability -

It is found by performing a series of experiments and observing their outcomes. These random experiments are also known as trial. The experimental probability for Event A can be calculated as -

$$P(E) = \frac{\text{Number of times event A happened}}{\text{Total no. of trials}}$$